



PRINTING MONKEY

Why printing money isn't the problem —
the problem is what it's invested in

Beken Bauher Marin

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Author's note

Every time a central bank announces it's going to "print money" — whether by buying bonds, funding a bailout, or backing a spending program — the almost automatic reflex from commentators, go-to economists, and financial outlets is the same: inflation alert, calls to raise interest rates, warnings that "stability is at risk." And they're almost always partly right.

But I've spent years watching something that reflex doesn't explain: two countries can print comparable amounts of money, in similar circumstances, and end up in completely different places. One builds new productive capacity, comes out of the crisis stronger, and absorbs that new monetary mass without destroying the value of its currency. The other spirals into lost confidence, shortages, and prices spinning out of control.

This book doesn't argue that "printing money is always good." That claim would be just as simplistic as the one it's trying to correct. What I argue is something more precise: the effect of creating new money depends almost entirely on what that money becomes before it reaches prices. If it becomes factories, infrastructure, energy capacity, innovation, or strategic materials, the real economy grows alongside the money supply and prices tend to stay stable. If it becomes nothing but consumption, transfers with no productive counterpart, or speculation, prices rise because the new money is chasing a supply of goods that didn't grow at the same pace.

I named this book *Printing Monkey*, not "*The Printing Press of the Future*," on purpose. *MonkeyMoney* was born to speak to young people for whom the traditional language of finance — neckties, gray charts, panels of experts — sounds like a boring class, not something that belongs to them. But the decisions discussed in this book do belong to them: the generation that's twenty or thirty years old today is going to live with the consequences of how governments and central banks handle money over the coming decades, so they'd better understand the rules of the game, even if no one ever explained them without the jargon.

Over the course of these pages, I'll first lay out the orthodox argument with all the seriousness it deserves — it's not a strawman, it's the foundation of monetary policy across almost the entire planet — and then I'll show where that theory falls short, drawing on economists and historians who have documented both sides of this coin: the cases where "printing" built the future, and the cases where it destroyed it completely.

My intention isn't for you to finish this book believing central banks should fund anything without limit. My intention is that the next time you hear the phrase "printing money is bad," your first question will be: bad compared to what, and compared to funding what instead?

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Introduction — The myth of the cursed printing press

Few phrases in economics get repeated as often as this one: "printing money causes inflation." It's taught in university, repeated on television every time a central bank announces a new bond-buying program, and used as a conversation-ending argument at dinner tables about why a government "can't just print more" to solve its problems.

The phrase isn't false. It's incomplete. It treats all money creation as identical, regardless of where it goes — as if pouring water into a full bucket or onto dry soil were exactly the same operation with the same result.

This book starts from a simple observation that orthodoxy rarely stops to explain in enough detail: when a central bank or a government creates new money, that money doesn't show up directly as higher prices at the supermarket. It first passes through a channel: it's lent to a company, it funds a public works project, it pays out a subsidy, it buys a bond, it capitalizes a development bank. What happens in that channel — whether it funds something that increases the quantity of goods and services available, or whether it simply funds more demand on top of the same old supply — determines whether the end result is growth or inflation.

Two countries, the same amount of money, two different endings

To keep this idea from staying abstract, picture two imaginary countries, A and B, facing the same crisis and each responding by creating the equivalent of 10% of their economy in new money. Country A puts it toward finishing a hydroelectric dam, modernizing its transportation network, and funding factories that expand production shifts. Country B puts it toward sustaining subsidies and current spending it had already committed to, with no investment plan behind it. Two years later, Country A has more electricity, more logistics capacity, more industrial employment, and barely noticeable inflation, because the supply of goods grew at the same pace as the money. Country B has exactly the same productive capacity as before, and inflation that keeps accelerating, because that new money can only compete for the same old goods.

That contrast, told with imaginary countries, repeats itself with real names and dates throughout the economic history of the last century, and this book is built, chapter by chapter, to show it to you with evidence, not faith.

The question this book wants to leave you with

The question this book poses isn't "should money be printed?" That question, framed that way, doesn't have a useful answer: every central bank in the world creates money every day as a normal part of its function. The right question is: to fund what, with what discipline, and with what capacity in the real economy to absorb it without prices rising?

We'll walk through that argument in three movements. First, we'll understand what money really is and why the orthodox quantity theory of money has a solid logic that deserves to be taken seriously. Second, we'll see where that theory has fallen short

against the evidence of the last two decades, drawing on economists who have documented those cracks from different angles. Third, and this is the heart of the book, we'll propose a concrete criterion — productive versus speculative — to distinguish the "printing" that builds a future from the kind that destroys the present, illustrated with the most extreme historical cases of both types: from the reconstruction of Europe after 1945 to the Hungarian hyperinflation of 1946, the worst ever recorded.

If, by closing this book, you find yourself looking at the next headline about a central bank with a sharper question than "how much did they print?", the book will have done its job.

Chapter 1 — What money really is

Before discussing whether creating money is good or bad, it's worth clarifying exactly what we're creating, because the popular image — a literal printing press churning out bills — describes how modern money works poorly.

From seashells to keyboards

For most of human history, money was a commodity: shells, salt, cattle, and eventually precious metals. Its value came, in theory, from being scarce and useful in itself. Over time, that commodity money gave way to representative money — bills that were a receipt redeemable for gold stored in a vault — and finally, during the twentieth century, almost every country moved to fiat money: bills not redeemable for anything physical, whose value depends entirely on the collective confidence that they'll be accepted to pay taxes and buy goods.

That transition wasn't an accident or a conspiracy: modern economies grew far faster than any gold mine on the planet. Tying the money supply to a specific metal meant, in practice, tying possible economic growth to geological luck.

Most money isn't printed: it's lent into existence

Here's the detail that most surprises anyone approaching this topic for the first time: in a modern economy, the vast majority of the money in circulation doesn't come out of the central bank's physical printing press. It's created when a commercial bank issues a loan. When a bank approves your loan to buy a house or start a business, it isn't handing out savings someone else left on deposit; it's creating a new deposit in your account, backed by the promise that you'll repay that loan with interest.

The central bank, for its part, doesn't "print" in the literal sense when it carries out what's known as quantitative easing or monetary financing: it adjusts electronic reserves in the banking system, buys government bonds or other assets, and injects liquidity that commercial banks can, in turn, lend out. The word "printing press" that we use in this book — including in the title — is a useful metaphor for monetary creation in general, not a literal description of the process.

An example with small numbers

Imagine a bank with 100 monetary units in deposits and a regulatory requirement to hold a fraction in reserve. When that bank lends 80 units to a company, those 80 units don't come from anywhere physical: they appear as a new deposit in the company's account, which it can then spend paying a supplier, who deposits it in another bank, which can lend out a fraction of it again. This process — known as the credit multiplier — is why the total amount of money in a modern economy is several times larger than the monetary base actually issued by the central bank. Monetary policy doesn't control that

final figure directly and mechanically; it influences it by adjusting interest rates, reserve requirements, and the amount of reserves it makes available to the banking system.

Money as social technology

Perhaps the most useful way to think about modern money isn't as a limited commodity that exists in a fixed quantity and gets divided up, but as a social technology: a system of collective bookkeeping that lets us coordinate who owes what to whom, and that lets one person's work turn into another person's good or service without a direct barter.

That perspective — shared, with different nuances, by Modern Monetary Theory and by a good deal of practical central banking — doesn't mean money can be created without limit or consequence. It means the relevant limit isn't "how much gold is in the vault," but how much real capacity the economy has to produce the goods and services that new money is going to want to buy. That's exactly the thread we'll follow through the rest of the book.

Chapter 2 — Orthodoxy and its logic

Before questioning a theory, you have to understand it in its best form. The idea that "printing money causes inflation" wasn't born out of nowhere: it's the conclusion of one of the most influential traditions in twentieth-century economics, and it has a solid internal logic that deserves respect.

The quantity theory of money

The modern version of this idea is associated above all with Milton Friedman, who summed it up in a phrase that became the banner of monetarism: inflation is always and everywhere a monetary phenomenon. In the work he wrote with Anna Schwartz, a monetary history of the United States covering nearly a century of data, Friedman and Schwartz documented how episodes of high inflation systematically coincided with periods of rapid money-supply expansion, and how poorly calibrated monetary contractions — like the one that, in their reading, deepened the Great Depression — could be just as damaging in the opposite direction.

The underlying logic is intuitive: if the amount of money in circulation grows faster than the quantity of goods and services available to buy with that money, the average price of those goods has to rise for the market to rebalance. More money chasing the same amount of bread can only end in more expensive bread.

Cagan and the technical definition of hyperinflation

Economist Phillip Cagan, in a classic paper edited by Friedman himself, proposed the definition still used today: a hyperinflation begins when monthly inflation exceeds 50% for at least several consecutive months. That definition matters because it draws a clear line between the uncomfortable but manageable inflation any economy can go through from time to time, and the absolute monetary collapse we'll study in Chapter 6: at 50% a month, a currency loses half its purchasing power every 30 days. At the rates reached by Germany, Zimbabwe, and Hungary at their worst, that loss happened in hours, not months.

Independent central banks and inflation targets

From that logic came the institutional design that has dominated monetary policy since the 1980s: central banks with formal independence from the government of the day, with an explicit mandate to keep inflation low and stable — typically around 2% a year — and with the interest rate as the main tool to cool the economy when prices accelerate.

The textbook case of this tradition is the disinflation carried out by Paul Volcker at the head of the U.S. Federal Reserve in the early 1980s. U.S. inflation had climbed past 13% a year by the late seventies. Volcker pushed the benchmark interest rate above 19%, deliberately triggered one of the harshest recessions of the postwar era — with unemployment climbing above 10% — and managed, by 1983, to bring inflation down to single digits. For the monetarist tradition, that episode confirmed that inflation is, above

all, a problem of monetary discipline, and that sustaining that discipline requires being willing to pay painful real costs in the short run.

The justified fear of hyperinflation

This tradition doesn't defend its limits out of theoretical whim: it defends them because there are devastating historical examples of governments that funded their spending by creating money with no anchor at all, and ended up completely destroying the value of their currency. We'll study some of those cases in detail — Hungary in 1946, the worst hyperinflation ever recorded; Germany in the 1920s; Zimbabwe; Venezuela — in Chapter 6, because they are real warnings, with real names, dates, and victims, not laboratory hypotheses.

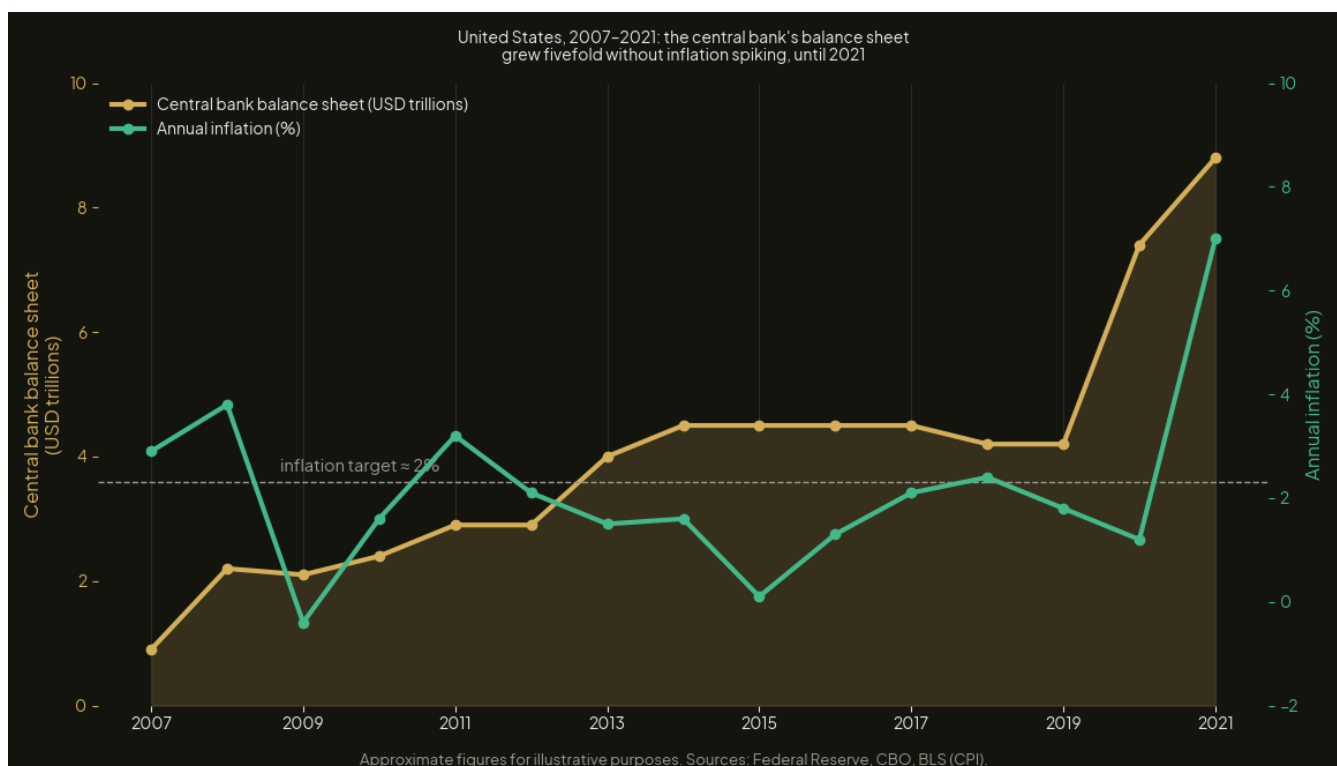
It's important to say this clearly before moving on: the orthodox tradition is right about something essential. Creating money with no limit or criterion at all, to fund anything whatsoever, eventually destroys a currency. Where this book departs from orthodoxy isn't in denying that possibility, but in questioning whether the only relevant criterion is "how much" is created, rather than "what for" it's created. That's exactly the ground we explore in the next chapter.

Chapter 3 — The cracks in the theory

If the quantity theory of money were the whole story, the years following the 2008 financial crisis should have been a decade of runaway inflation in the United States, the eurozone, the United Kingdom, and Japan. The central banks of those economies created, through quantitative easing programs, unprecedented amounts of money in peacetime to buy government bonds and other financial assets.

The inflation that never came

The U.S. Federal Reserve's balance sheet went from about \$0.9 trillion in 2007 to about \$4.5 trillion by 2014–2015 — a fivefold increase — through three successive rounds of quantitative easing known as QE1, QE2, and QE3. If the quantity of money alone were the reliable predictor of inflation that the simplest version of quantity theory suggests, that expansion should have sent prices soaring. Instead, U.S. inflation stayed below the central bank's 2% target for almost that entire period, and in 2015 it dipped to around just 0.1% — so low that the dominant worry in monetary policy circles at the time wasn't the risk of inflation, but the opposite risk: deflation.



The U.S. central bank's balance sheet grew fivefold between 2007 and 2015 without inflation coming anywhere near worrying levels. It only spiked again in 2021, alongside a second jump in the balance sheet and a severe supply disruption.

Economic historian Adam Tooze, in his analysis of the global financial crisis and its aftermath, documents how that massive monetary expansion largely stayed within the financial system — pushing up prices of assets like stocks and bonds — without transmitting with the same force to the prices of goods and services that traditional inflation measures.

That anomaly doesn't invalidate quantity theory, but it does reveal something the simplest version of the theory doesn't explain well: the channel through which money is created matters as much as the amount. Money that stays trapped in financial assets, bought by investors who already had plenty of liquidity, doesn't compete for bread or gasoline the same way money that lands directly in the pocket of someone who's going to spend it right away does.

Modern Monetary Theory and the limit of real resources

That observation is one of the central points of Modern Monetary Theory, popularized by economists like Stephanie Kelton in her book *The Deficit Myth*. Her central argument is that a country that issues its own sovereign currency doesn't face, strictly speaking, a financing limit the way a household or a business would: the real limit is the economy's productive capacity. It can create whatever money it needs to pay for things denominated in its own currency; what it can't create are the workers, factories, raw materials, or energy needed to produce what that money wants to buy. When spending funded by new money runs up against those real limits — when there's no more idle capacity left to activate — that's when inflation shows up, and not before.

That idea, stripped of its more controversial packaging, isn't so different from an argument John Maynard Keynes had made decades earlier in his *General Theory of Employment, Interest and Money*: in an economy with idle productive capacity — shut-down factories, unemployed workers — injecting additional spending, funded with new money if necessary, activates that capacity without generating significant inflationary pressure, because supply can respond. The problem only shows up once you keep injecting spending after that idle capacity has already been used up.

The 2021–2023 inflation: an uncomfortable test for both sides

The inflationary surge that followed the COVID-19 pandemic complicates any overly tidy narrative. Between the last quarter of 2019 and the fourth quarter of 2021, the Federal Reserve's balance sheet doubled again, this time from about \$4.2 to about \$8.8 trillion, in a faster expansion than any previous round. U.S. inflation, which had spent more than a decade below 2%, climbed above 8% a year by 2022, its highest level in four decades.

But unlike the decade after 2008, this time there were, simultaneously, supply chains broken by lockdowns, a war that sent energy and food prices soaring, and production bottlenecks — from semiconductors to shipping containers — that had nothing to do with the amount of money in circulation. The academic debate over how much of that inflation came from the demand side — new money chasing goods — and how much from the supply side — goods that were scarce regardless of the money available — remains open, and this book doesn't claim to settle it.

What that episode does make clear, just like the decade after 2008, is that treating all money creation as equivalent — without distinguishing whether it funds new productive capacity, consumption against a fixed supply,¹³ or already-existing financial assets — leaves out the variable that best explains why some episodes of monetary expansion end in runaway inflation and others don't. That variable is the subject of the next chapter, and it's the central argument of this book.

Chapter 4 — The decisive criterion: productive vs. speculative

If the previous chapters made one thing clear, it's that the question "how much money was created?" isn't enough to predict whether the result will be growth or inflation. A second question is needed, and this book proposes that it's the one that truly decides the outcome: what did that money turn into before it reached prices?

Two possible destinations for new money

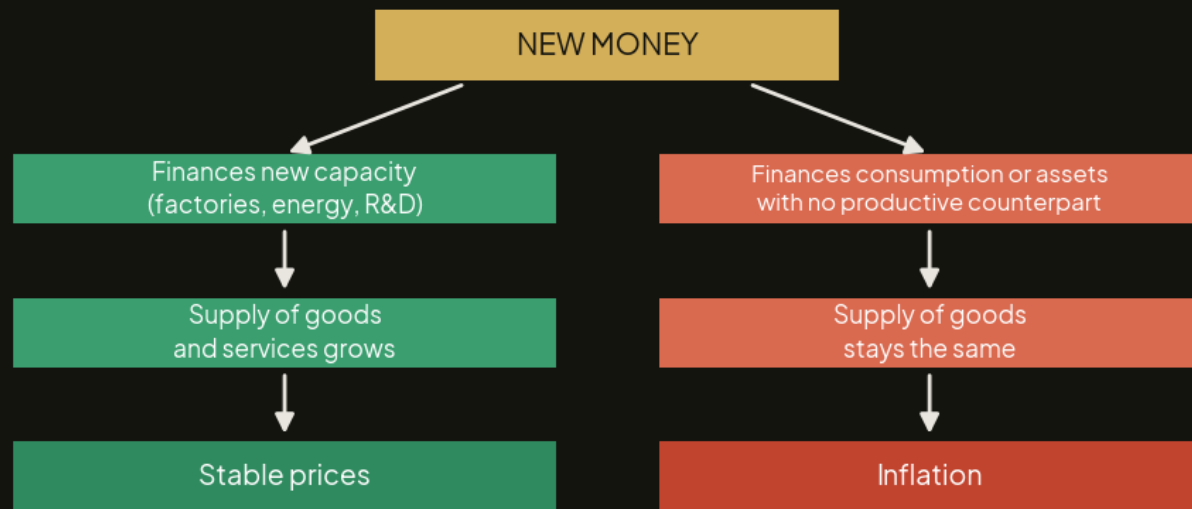
Consider two governments, each creating the equivalent of 5% of their economy in new money during a year of crisis.

The first channels it, through a development bank or a public works program, into building a more efficient power grid, modernizing ports, funding companies that expand their manufacturing capacity, and granting research funding in energy or agriculture. A year later, that economy doesn't just have more money circulating: it has more electricity available, more logistics capacity, more factories running, and new technologies that didn't exist before. The supply of goods and services grew alongside the money. Prices, on the whole, stay relatively stable because the new supply absorbs the new demand.

The second channels the same amount of new money into direct transfers with no investment counterpart at all, or into funding a deficit that covers the same current spending year after year, with no change to the country's productive capacity. A year later, that economy has more money circulating, but exactly the same amount of goods and services as before. That new money can only compete for the existing supply, and the result, almost inevitably, is that prices rise to ration that supply among more money chasing it.

The difference between the two scenarios isn't the amount of money created. It's the destination.

Two paths for new money



The same amount of new money can end up as stable prices or as inflation, depending on the path it takes before reaching the consumer.

Absorptive capacity as the real limit

This criterion has a useful technical name: an economy's absorptive capacity — that is, how much additional investment it can effectively convert into more real output before running into bottlenecks: a lack of skilled labor, materials, energy, logistics infrastructure. A well-designed investment program doesn't just pick productive projects: it sizes them so they don't exceed that absorptive capacity, because even money aimed at a productive purpose can generate inflation if it's injected faster than the economy can turn it into physical output.

The state as a risk investor, not just a crisis manager

Economist Mariana Mazzucato, in *The Entrepreneurial State*, documents in detail a point the popular image of the public sector as a mere "regulator" or "crisis firefighter" tends to ignore: much of the technological innovation later capitalized on by the private sector originated in direct public investment, largely funded with money created or guaranteed by the state, at a stage too uncertain and too far from any return for private capital to fund on its own. GPS, touchscreens, the protocols that gave rise to the internet, and a good deal of the pharmaceutical research behind key medications all trace their origins to this kind of state funding, through agencies like DARPA in the United States.

Mazzucato doesn't argue that any public spending funded with new money is automatically productive — she herself distinguishes between investment aimed at creating new capacity and spending that simply maintains the status quo — but her work is one of the strongest and best-documented defenses of this book's central idea: the problem was never that the money came from the state or from a printing press. The problem is whether that money funds the creation of something new or simply redistributes what already existed.

Chapter 5 — When it worked: the press that built the future

The economic history of the last century has several episodes where large-scale money or credit creation, explicitly aimed at expanding productive capacity, coincided with decades of growth without the inflation crises the simpler version of orthodoxy would predict. None of these cases is an unqualified utopia — all of them have legitimate criticisms worth mentioning — but they all share the same underlying pattern: new money, channeled with institutional discipline, into new productive capacity.

The U.S. Reconstruction Finance Corporation

During the Great Depression and World War II, the U.S. government operated a public lending institution, the Reconstruction Finance Corporation, which funded — with capital ultimately backed by the Treasury and the state's power to issue money — banks, railroads, and above all the enormous expansion of industrial capacity needed for wartime production. That industrial capacity, funded with public credit, wasn't dismantled once the war ended: it became a large part of the manufacturing base that sustained U.S. growth for the following three decades.

The 1948 German monetary reform and the Marshall Plan

Few cases illustrate this book's contrast better than Germany itself, comparing 1923 to 1948. On June 21, 1948, the Western occupation authorities replaced the devalued Reichsmark overnight with a new currency, the Deutsche Mark, in a reform designed to restore monetary confidence from the ground up. That reform coincided with the arrival of Marshall Plan funds: between 1948 and 1952, the United States financed the reconstruction of Western Europe with an aid program that, relative to the size of the era's economies, represented an extraordinary transfer of capital.

The key to its design wasn't just the amount, but the condition attached to it: the funds were tied to specific industrial and infrastructure reconstruction plans in each recipient country, with tracking mechanisms over how the money was invested. The result, in postwar West Germany, was the period of accelerated growth Germans themselves named the *Wirtschaftswunder*, the "economic miracle": the same country that, twenty-five years earlier, had lived through the most studied hyperinflation in the world became, in less than a decade, one of the strongest industrial economies in Europe. The difference wasn't that less money or credit was created in 1948 than in 1923 in relative terms: it was that this time it was tied, from the outset, to a currency with credible institutional backing and to concrete productive reconstruction plans, not to the desperate financing of war reparations.

East Asia's state-directed development economies

Economist Ha-Joon Chang, in *23 Things They Don't Tell You About Capitalism* and in his historical work on the industrial policies of today's developed countries, documents how

South Korea, Taiwan, and Japan, at different stages of their postwar development, used state development banks and directed credit — in practice, deliberate decisions about which sectors new financing should flow to — to build, in a matter of decades, industrial capacity in sectors like steel, shipbuilding, and electronics that no private financial market, left to its own short-term judgment, would have funded at the same scale or with the same patience.

Chang is emphatic that this directed credit came with strict performance conditions attached — companies receiving preferential financing had to meet export and productivity targets, and those who didn't lost access to that credit in later rounds — which is exactly the kind of institutional discipline that separates these cases from simply handing out money with no strings attached.

China's infrastructure over the last two decades

The expansion of directed credit toward infrastructure in China since the early 2000s — roads, high-speed rail, power generation capacity, ports — is, arguably, the most visible contemporary example of productive capacity built at an unprecedented scale and speed, largely funded by state banks operating under industrial-policy guidance. It's also a case worth mentioning honestly about its risks: that same model has generated levels of corporate and municipal debt that worry Chinese regulators themselves, and sectors like real estate today show the symptoms of having received credit beyond what real demand could justify. It's, in itself, a reminder that even credit directed toward productive capacity needs limits and discipline: the line between "productive investment" and "speculative overinvestment" can be crossed even within a single program.

Operation Warp Speed: funding capacity before approved demand existed

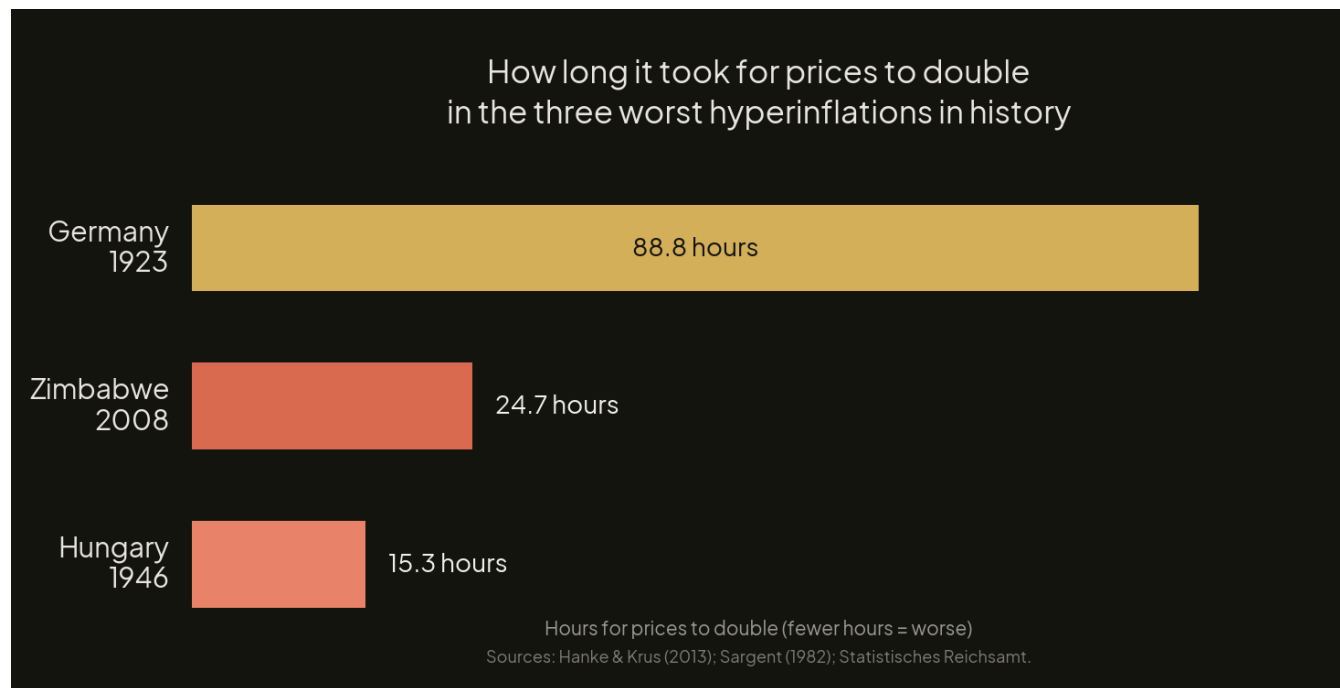
We already mentioned this in the previous chapter, but it deserves its place here among the cases that worked: in 2020, the U.S. government committed billions of dollars in public funds to finance, in advance, the construction of manufacturing plants for several COVID-19 vaccine candidates, before any of them had completed clinical trials. The institutional design included a key element that connects with the rest of the cases in this chapter: the money wasn't handed out as a generic subsidy, but tied to specific production and delivery contracts, with verifiable volume and timeline targets. When the vaccines turned out to be effective, that manufacturing capacity was already in place to produce hundreds of millions of doses within months, not years. The cost of funding capacity that ended up unused — because some candidate vaccines failed — was real, but small compared to the value of having capacity ready once the science confirmed which candidates worked.

The common thread

Across all five cases there's a shared pattern: the new money or credit wasn't distributed indiscriminately, but through institutions with an explicit mandate to fund productive capacity — a public lending corporation, a monetary reform tied to reconstruction plans, a development bank with performance targets, a manufacturing-contracts program with verifiable deliverables — and with some mechanism, more or less imperfect, to cut off financing when it stopped serving its purpose. That institutional design matters as much

Chapter 6 — When it failed: warnings worth taking seriously

No argument in favor of well-directed money creation is honest if it doesn't look squarely at the cases where printing money ended in catastrophe. The four episodes in this chapter aren't minor anecdotes: they are the real, justified reasons behind orthodoxy's fear of runaway inflation.



The three worst hyperinflations recorded with precision in history. Hungary 1946 remains, by a wide margin, the most extreme ever measured.

Hungary, 1945–1946: the worst hyperinflation ever recorded

Few people outside economic history circles know this case, despite it being, by the numbers, the most extreme in all of documented monetary history. After the devastation of World War II — the siege of Budapest alone left tens of thousands of civilian dead and destroyed much of the country's productive capacity — the Hungarian government funded reconstruction and its postwar obligations by issuing growing amounts of pengő, the national currency. By July 1946, monthly inflation reached a figure that can barely be written without scientific notation: 41.9 quintillion percent in a single month, with prices doubling, on average, every fifteen hours.

The government went as far as issuing a hundred-trillion-pengő note — a one followed by twenty zeros — and printed, though it never entered circulation, a thousand-trillion note. The amount of paper needed to print those astronomical denominations was so large that it caused a paper shortage that, ironically, delayed the printing of the new currency — the forint — that finally ended the crisis on August 1, 1946. By then, the accumulated value of every pengő in circulation amounted to a fraction of a U.S. cent.

Germany, 1921–1923

After World War I, the Treaty of Versailles imposed reparations obligations on Germany denominated in foreign currencies and gold, which the German government couldn't simply "create" the way it could create marks. To obtain that foreign currency and sustain public spending, the government resorted to issuing growing amounts of marks to buy it on the market, in a context where the country's productive capacity — damaged by the war and, from 1923 on, by the Franco-Belgian occupation of the industrial Ruhr region — wasn't growing at all.

The figure that best sums up the collapse is the exchange rate: a dollar cost seventeen thousand marks in January 1923; by November of that same year, it cost 4.2 trillion marks. In October 1923, monthly inflation reached around 29,500%, with prices doubling every 3.7 days. A loaf of bread that cost one hundred sixty marks at the end of 1922 came to cost two hundred billion marks by the end of the crisis.

The pattern is the exact opposite of the cases in the previous chapter: new money created to cover obligations and current spending, with no connection to expanding productive capacity, in an economy whose real supply was, on top of that, shrinking.

Zimbabwe, the 2000s

Zimbabwe's case adds a crucial element: runaway money creation coincided with the active destruction of the country's main source of production. The land reform program of the early 2000s displaced most of the commercial farmers sustaining export production, and agricultural output — the backbone of the economy — collapsed. The government, facing a growing fiscal deficit, a war in the Congo, and an ever-shrinking economy, funded public spending by issuing money at a pace that drove annual inflation from an already-high 55% in 2000 to more than 2,000% by 2007, peaking in November 2008: an estimated monthly inflation of 79,600,000,000%, an annualized equivalent of 89.7 sextillion percent, and prices doubling every 24.7 hours. The central bank went so far as to issue a hundred-trillion-Zimbabwean-dollar note that, at the peak of the devaluation, wasn't enough to pay for a bus fare.

Venezuela, the 2010s

The Venezuelan case combines both of the previous elements with a third: extreme dependence on a single source of income. The decline in oil production — from lack of investment, maintenance, and technical management at the state oil company — drastically reduced the country's main source of foreign currency, right as international oil prices also collapsed. Venezuela's central bank funded the resulting deficit through money creation on a scale that drove annual inflation from 274% in 2016 to 863% in 2017 and an estimated 130,060% in 2018, with cumulative inflation since 2016 that the authorities themselves put, by 2019, at more than 53 million percent. That hyperinflation, combined with a supply of basic goods that depended heavily on imports that grew ever harder to pay for, destroyed the country's savings, wages, and price system for several years, and pushed millions of people to emigrate.

Chapter 7 — Material backing: do we need to go back to gold?

Faced with the cases in the previous chapter, an understandable reaction is to think the solution is to tie money to something physical and limited — gold, silver, a basket of commodities — so that no government can create more than that backing allows. This idea has a long history, mixed results, and it's worth understanding before proposing an alternative.

The gold standard and its end

For much of the nineteenth and twentieth centuries, the world's major currencies were tied, with varying degrees of rigidity, to gold. The Bretton Woods system, agreed in 1944, established an international monetary order where the U.S. dollar was convertible to gold at a fixed rate of \$35 an ounce, and other major currencies were pegged to the dollar. That system worked reasonably well during postwar reconstruction, but by 1971 U.S. gold reserves were no longer enough to back the amount of dollars circulating in the rest of the world, and President Nixon unilaterally suspended convertibility, an episode known as the "Nixon shock" that marked the effective end of the international gold standard.

Why even the monetarists moved away from gold

It's revealing that Milton Friedman, the economist most associated with the warning that creating too much money causes inflation, was also an explicit critic of the gold standard as a solution. In *Free to Choose*, Friedman argues that tying the money supply to the stock of a specific metal doesn't eliminate the possibility of monetary crises: it simply relocates them elsewhere — an unexpected gold discovery, a drought in new mining output — and, worse, it strips the central bank of the ability to respond flexibly to genuine demand shocks, like a severe recession where the economy needs more liquidity, not less. His alternative proposal wasn't gold, but a predictable, moderate money-growth rule, actively managed by a responsible monetary authority.

A different idea: dynamic backing in real capacity

The proposal this book defends isn't a return to a fixed metal, but a reinterpretation of the concept of "backing": instead of tying money creation to a fixed quantity of a single material, tie it to the verifiable expansion of real productive capacity — energy, installed industrial capacity, strategic reserves of materials key to future sectors like the minerals the energy transition requires, logistics infrastructure. The difference from the gold standard is that this backing can grow alongside the real economy, instead of staying fixed to the geological luck of a single element, while preserving the discipline the gold standard sought to impose: no new money is created unless, on the other side, something of equivalent real value exists or is verifiably under construction.

This idea isn't entirely new — some development banks and sovereign natural-resource funds already operate, in practice, under a similar logic when they tie financing to

specific reserves or productive projects — but it's rarely discussed as a general principle of monetary design. The next chapter tries to turn this idea, together with everything we've seen so far, into a concrete set of criteria.

Chapter 8 — Designing the printing press of the future

Everything above — orthodoxy, its cracks, the cases that worked and the ones that failed, the idea of dynamic backing — can be condensed into a practical set of criteria. It isn't a mathematical formula that guarantees the outcome; it's a list of questions any program of money creation or directed credit should be able to answer satisfactorily before it's implemented.

Six questions before creating new money

- Does this financing expand the economy's real productive capacity, or does it only add demand on top of supply that already exists?
- Is there an institution with a mandate and verifiable performance targets in charge of channeling it, rather than a discretionary disbursement?
- Does the pace of the injection respect the receiving sector's absorptive capacity — labor, materials, available infrastructure?
- Is there a review mechanism and an eventual exit, rather than perpetual financing with no evaluation?
- Is success measured in real goods and services produced, not just in nominal spending or GDP-growth figures?
- Is there public transparency about who receives the financing and under what conditions?

The table below summarizes the contrast between meeting and failing each criterion, with examples of the type already discussed in previous chapters.

Criterion	Example that meets it	Example that fails it
Does it expand real productive capacity, not just demand?	A development bank funds a new plant.	A transfer with no investment counterpart.
Is there institutional tracking with performance targets?	Directed credit with export conditions.	Discretionary disbursement with no conditions.
Does it respect the sector's absorptive capacity?	Phased investment based on available labor.	Stimulus that exceeds available capacity.
Is there a wind-down plan once the project matures?	Financing with a review and exit date.	Permanent financing with no review.
Is it measured in real goods and services, not just nominal figures?	Installed capacity is reported, not just spending.	Success measured only in money spent.
Is there public transparency about who and why financing is granted?	Open records of recipients and targets.	Financing decisions with no public record.

A framework, not a guarantee

It's worth being honest about the limits of this framework. No design criterion fully eliminates the risk of corruption, political capture, or simple miscalculation of how much absorptive capacity a sector really has. Development banks can turn into instruments of political favoritism; infrastructure projects can be overbuilt, as the Chinese case mentioned in Chapter 5 itself shows. This framework doesn't claim to be a mathematical guarantee; it aims to be a better question than the one we ask today, which is usually limited to a single number — how much will be spent — without ever asking toward what.

Chapter 9 — The role of tomorrow's central banks

If we accept that where new money goes matters as much as how much of it there is, an uncomfortable institutional question follows: should central banks' mandate expand beyond the inflation target, to explicitly incorporate some criterion about the quality or destination of the credit their policy enables?

Mandates already starting to move in that direction

This isn't an idea entirely foreign to current practice. Some central banks have begun exploring what monetary-policy circles discuss as directed credit with green or climate criteria: more favorable financing or reserve conditions for activities that reduce emissions, compared with those that increase them. The Bank of Japan, for much of its postwar history, coordinated closely with the industry ministry to channel credit toward strategic sectors, in an institutionalized version of the same principle we saw in Chapter 5. None of these examples is a perfect solution, nor free of debate, but they show that expanding monetary policy's criteria beyond a single number isn't a theoretical fantasy: it's already happening, partially, in several places.

The risk that can't be ignored: politicization

Central bank independence from the government of the day isn't a technocratic whim: it exists precisely to prevent money creation from being used to fund short-term election spending instead of long-term goals. Any proposal to expand a central bank's mandate toward "credit quality" criteria runs a real risk of opening the door to political pressure to use that criterion, in practice, to favor sectors with more political influence rather than genuinely greater productive capacity.

This is, perhaps, the hardest balance in the whole book: the very argument that justifies expanding the criterion beyond "how much" toward "what for" is the one that can be hijacked by special interests if the institution applying that criterion doesn't have enough technical independence and public accountability. The answer isn't to avoid the question of where credit goes out of fear of that risk, but to design institutions — like the development banks with verifiable performance targets we saw in Chapter 5 — with enough transparency and technical autonomy to resist that capture, knowing that no institutional design eliminates it entirely.

Chapter 10 — Why this matters to your generation

If you've made it this far and the word "macroeconomics" still sounds like a boring class that has nothing to do with your life, this chapter is for you. Because it has nothing to do with a class. It has to do with your income, your salary, whether you'll ever be able to buy something of your own, and who gets to decide that without asking your permission.

You already live inside these decisions

Every time a central bank decides how much money to create and where to direct it, it's indirectly deciding how expensive your life is going to be next year, whether landing your first job is going to be easy or brutal, and whether the money you manage to save is going to hold its value or evaporate while you weren't looking. This isn't a topic for adults in neckties in some distant boardroom: it's, literally, the topic that most defines how much your adult life is going to cost you.

The generation that's between 15 and 35 today reached adulthood — or is reaching it — after two of the largest monetary experiments in history: the response to the 2008 crisis and the response to the 2020 pandemic. You watched, without anyone explaining it clearly, trillions of dollars, euros, and other currencies get created within months. You lived through, firsthand, the inflation that followed the second round. You weren't a spectator to this book: you were part of the experiment.

You don't need an economics degree to have an informed opinion

One of the most effective ways to keep people out of a conversation is to make the language sound impossible to understand without years of study. Monetary economics has its share of real technical complexity, but this book's central question — what is this money being created for, and how likely is it that the economy can produce more with it? — doesn't need a doctorate to ask. It needs the habit of asking it, instead of accepting the headline at face value.

Monkey Money exists exactly for that: so you can practice these decisions through play, with no one charging you for getting it wrong, before you have to make them with real money. The monkey on this book's cover doesn't wear a suit because the decisions this book talks about shouldn't stay locked inside a suit. They belong to you as much as to any central banker.

What you can do with this, starting today

- Next time you see a headline about a central bank "printing money," look up what it says the money is for — don't stop at the amount.
- When a government announces a large spending program, ask yourself whether it sounds like investment with measurable targets or a transfer with no conditions.
- Practice this book's criterion in MonkeyMoney's games before applying it to real decisions with your own money.

- Share this way of thinking with anyone who repeats the phrase "printing money is always bad" with no further context. They probably were never told the other half of the story.

That is, at bottom, the real "printing press of the future" this book talks about: not a machine, but a mental habit your generation can make normal before the previous one did.

Conclusion — The question that really matters

We began this book with a phrase repeated so often it's stopped being examined: printing money causes inflation. Throughout these pages we haven't denied it — it can't be denied, it's backed by real, devastating cases like Hungary in 1946, Germany in 1923, Zimbabwe, and Venezuela, which deserved all the seriousness we gave them in Chapter 6 — but we've put it in its proper place: it's a partial truth that describes what happens when new money doesn't meet, on the other side, more real productive capacity to absorb it.

The question that truly decides the outcome was never "how much?" It was always "for what, with what institutional discipline, and respecting what real absorptive capacity?" A development bank that funds new industrial capacity with verifiable performance targets — as in Germany in 1948, in South Korea, or in Operation Warp Speed in 2020 — and a government that covers current spending indefinitely with no productive counterpart at all — as in Hungary in 1946 or in Venezuela over the last decade — can, in theory, create exactly the same amount of money. The results, as we've seen again and again in this book, have nothing in common.

That distinction — productive versus speculative, new capacity versus redistribution over a fixed supply, institutional discipline versus discretion with no conditions attached — is the printing press of the future this book refers to. It isn't a press that never stops, nor one that's never used. It's one that asks, before it starts running, what it's going to build.

Whether you work in public policy, at a bank, at a company, whether you're just starting your financial life, or whether you simply follow economic news as someone who wants to understand the world you live in, I invite you to take one single tool away from this book: next time you hear that a government or a central bank is going to "print money," before getting alarmed or excited, ask what for. That question, more than any inflation figure, is the one that truly matters.

Recommended reading

This book converses with — and sometimes disagrees with — the following works. Reading them directly is recommended for going deeper into each tradition:

- Stephanie Kelton, *The Deficit Myth*, 2020.
- Mariana Mazzucato, *The Entrepreneurial State*, 2013.
- Ha-Joon Chang, *23 Things They Don't Tell You About Capitalism*, 2010.
- Ha-Joon Chang, *Kicking Away the Ladder*, 2002.
- Milton Friedman and Anna Schwartz, *A Monetary History of the United States, 1867–1960*, 1963.
- Milton Friedman, *Free to Choose*, 1980.
- John Maynard Keynes, *The General Theory of Employment, Interest and Money*, 1936.
- Adam Tooze, *Crashed: How a Decade of Financial Crises Changed the World*, 2018.
- Steve H. Hanke and Nicholas Krus, *World Hyperinflations*, 2013 (reference table of hyperinflation episodes).

Beken Bauher Marin writes from MonkeyMoney, a space for financial education to play, read, and talk about money without unnecessary jargon.

PRINTING MONKEY

"Printing money is bad." You've heard it a thousand times, almost always without anyone explaining the other half of the story: that the same money, directed toward building real productive capacity, has funded reconstructions, vaccines, and entire infrastructures without destroying a single currency — and that the same tool, badly directed, has wiped out the savings of entire countries in a matter of weeks.

With cases ranging from the 1946 Hungarian hyperinflation — the worst ever recorded — to the 2020 Operation Warp Speed, Beken Bauher Marin builds a simple, applicable criterion for telling apart a printing press that builds a future from one that destroys the present: it isn't how much is created, it's what for.

A MonkeyMoney book, written for those who run from finance but can't run from its consequences.

MonkeyMoney — financial education to play, read, and talk about.